All the psvm and images can be found here inside the HW6 folder:
https://github.com/elizaan/Viz-Scientific-Data-HWs

Part 1

1.

Ans:

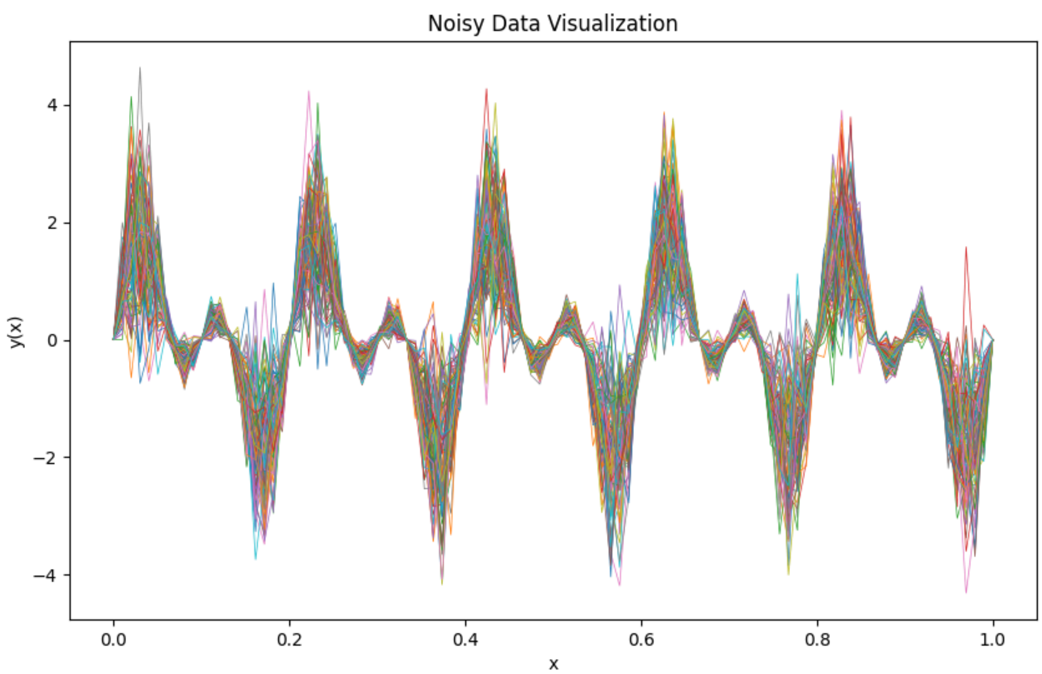


Figure 1: Visualization of Noisy Data

Figure 1 shows 100 different instances of the noisy signal plotted simultaneously. This creates a "spaghetti plot" that visually represents the uncertainty in the data. This approach creates heteroscedastic noise, where the amount of noise varies with the signal magnitude - higher amplitude regions experience more noise than regions near zero. This effectively models many real-world measurement scenarios where measurement error is proportional to the signal strength. I learnt the following things:

- The underlying sinusoidal pattern remains visible despite the noise

- The spread of lines is wider at peaks and troughs, demonstrating how the noise is proportional to signal magnitude
- The zero-crossing regions show less variability, making these areas more certain
- The overlapping of many instances creates a visual density that intuitively conveys probability

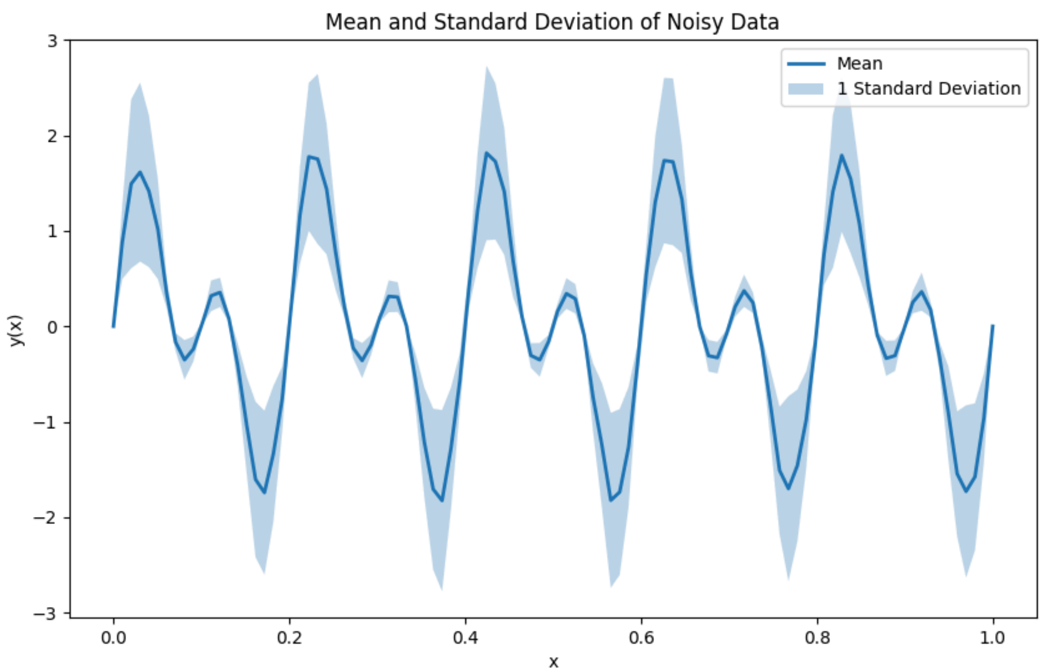


Figure 2: Statistical Representation of Uncertainty

Figure 2 shows:

- The mean curve (blue line) - representing the central tendency of all noisy instances
- The shaded region, representing one standard deviation above and below the mean

This representation effectively communicates:

- The most likely value at each point (the mean)
- The uncertainty around that value (the standard deviation band)
- How uncertainty varies across the domain (wider bands at peaks/troughs)

The heteroscedastic nature of the noise is visible in the varying width of the standard deviation band.

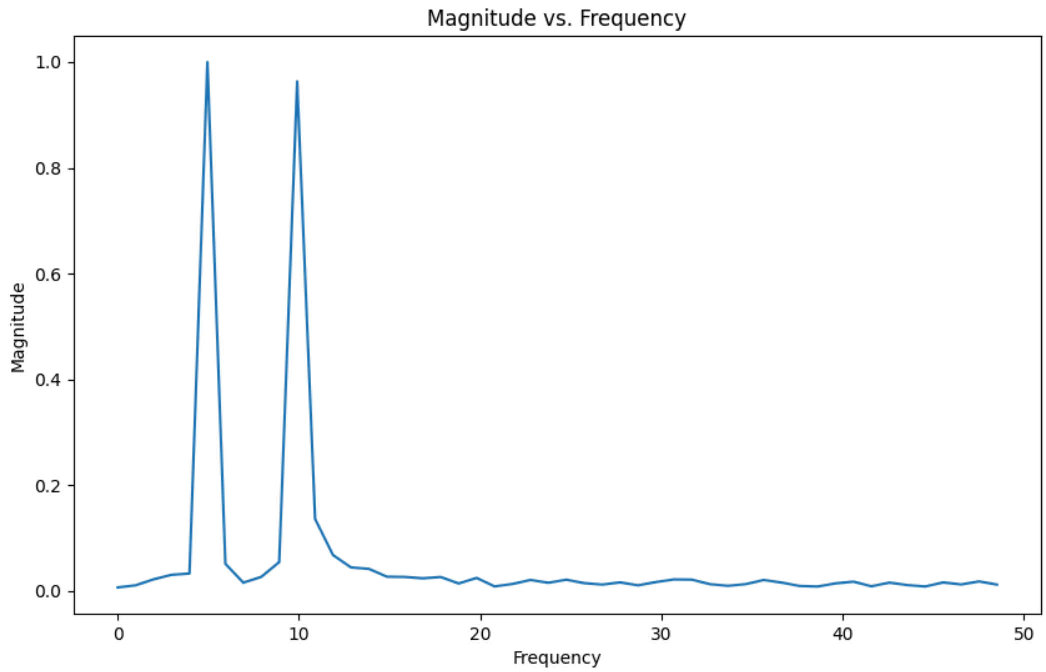


Figure 3: Frequency Analysis

We applied a Fast Fourier Transform (FFT) to the mean curve to better understand the signal composition and prepare for noise removal. The frequency domain representation shows (Figure 3):

- Two prominent peaks representing the two sine components of the original signal:
  - A peak at approximately 5 Hz (corresponding to $\sin(10\pi x)$)
  - A second peak at approximately 10 Hz (corresponding to $\sin(20\pi x)$)
- Low-amplitude noise spread across other frequencies

This spectral analysis confirms that despite the added noise, the fundamental frequency components of the original signal remain dominant and potentially recoverable through appropriate filtering.

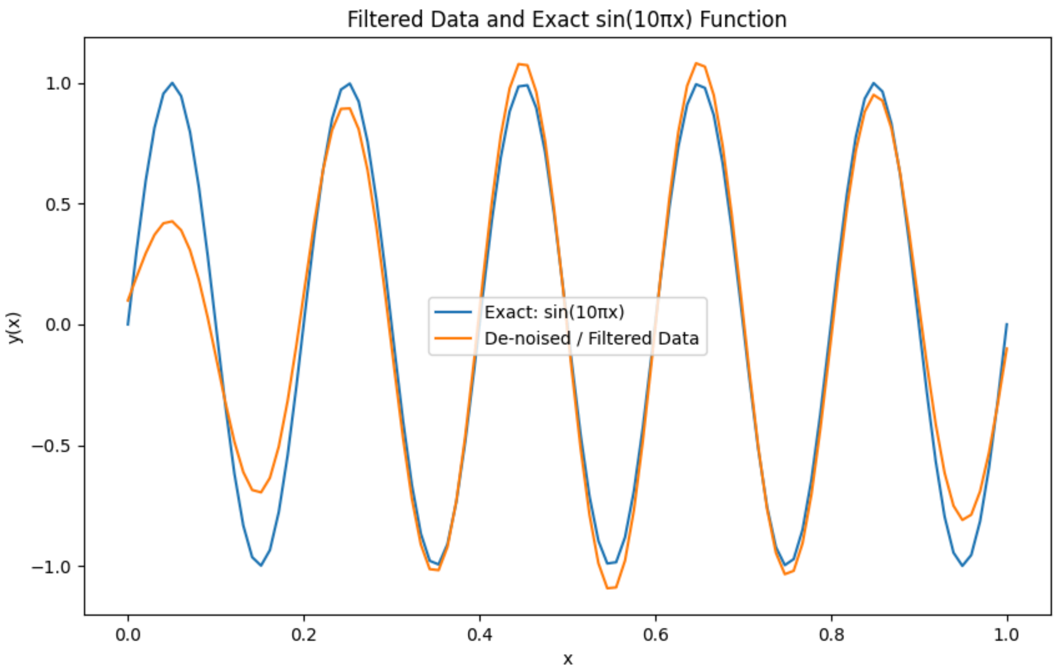


Figure 4: Signal Recovery through Filtering

We applied a bandpass Butterworth filter to isolate and recover the 5 Hz component (sin(10πx)) from the noisy data. The filter parameters were (Figure 4):

- Filter type: Bandpass Butterworth
- Filter order: 2
- Passband: 4-6 Hz

The filtered result (orange line) is compared with the exact sin(10πx) function (blue line). Observations:

- The filter successfully recovered the sin(10πx) component from the noisy data
- The recovered signal closely matches the exact function for most of the domain
- Some discrepancies are visible, particularly at lower x values
- The higher frequency component (sin(20πx)) has been effectively removed

The difference at lower x values can be attributed to edge effects in the filtering process. Butterworth filters can introduce phase distortions, especially at the boundaries of the data. These edge effects are a common challenge in signal processing and are more pronounced at the beginning of the signal.

2.

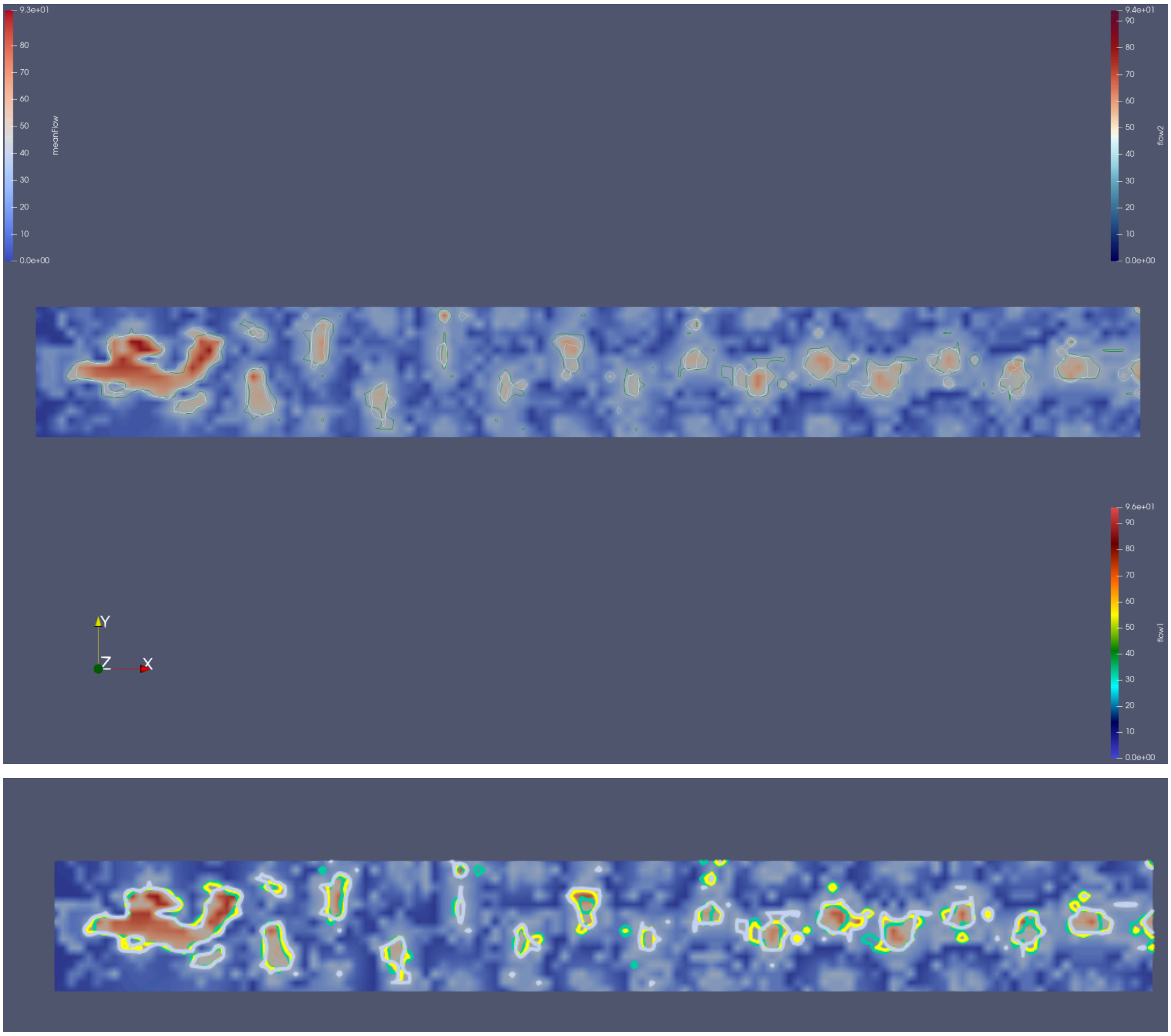
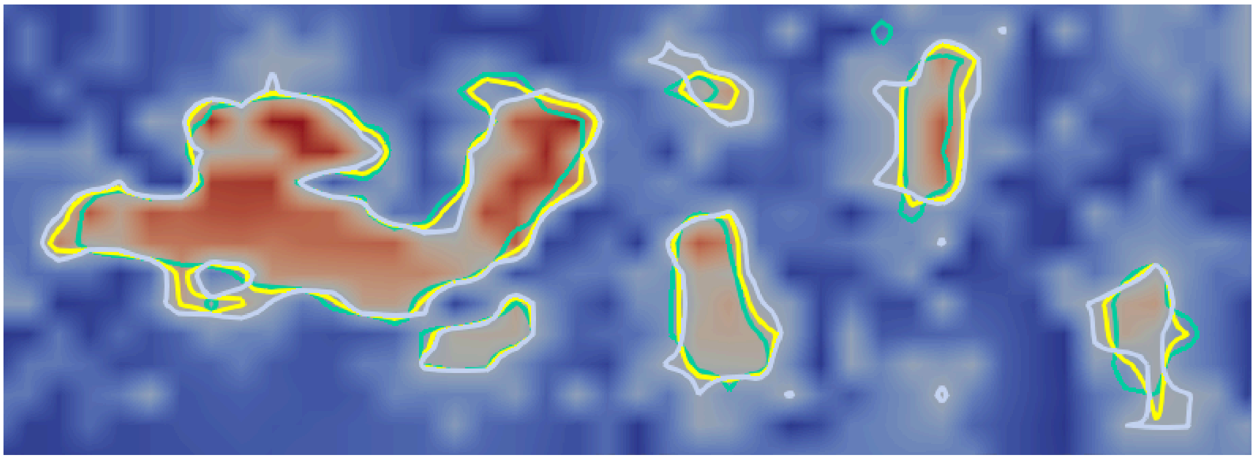


Figure 5: Isocontour value 4 applied to all data

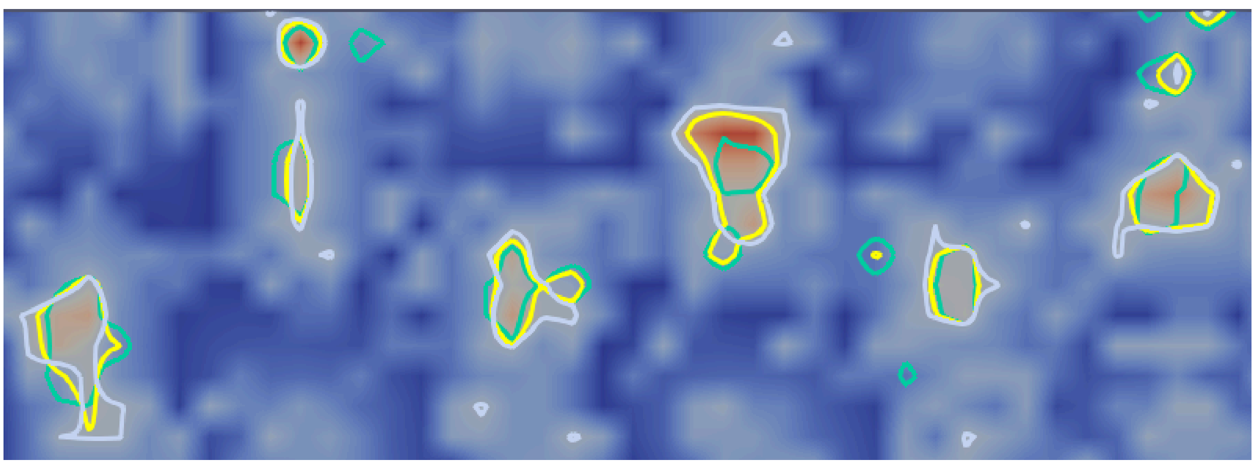
Figure 5 shows that the isocontours do not entirely coincide across the three flow datasets. This lack of coincidence can be attributed to several factors:

<u>Viscosity Effects:</u> The primary reason for the differences is that flow1.vtk and flow2.vtk represents the same flow captured with different fluid viscosity values. Viscosity directly affects flow behavior - higher viscosity creates more resistance to flow, resulting in different flow patterns and velocity distributions.
<u>Boundary Layer Effects:</u> In regions near boundaries or obstacles, the different viscosity values lead to varying boundary layer thicknesses and separation points, creating notable differences in isocontour positions.

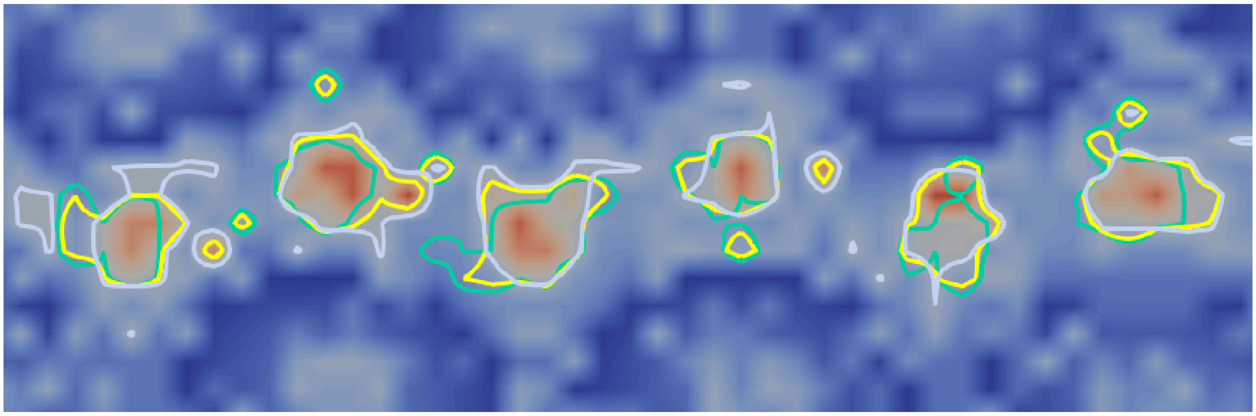
Vortex Formation: Areas of vorticity and recirculation are susceptible to viscosity changes, leading to significant positional variability in the isocontours in these regions.
Mean Flow Representation: The meanFlow.vtk creates an average of the two viscosity conditions, creating an isocontour that doesn't perfectly match the individual flow simulations.



(a)



(b)



(c)

Figure 6: Zoomed-in Values of different parts for Isocontours

From Figure 6, I can identify several regions with distinct patterns of positional variability:
In Figure 6(a), High Variability (Upper Left) is visible in the large flow structure, particularly along the irregular edges and protrusions. This suggests this area contains flow structures susceptible to viscosity changes, possibly due to complex flow separation or vortex shedding. In Figures 6(b) and 6(c), High Variability (Center) in the center structures shows considerable differences in shape and position between the different viscosity conditions. The boundaries of these structures do not closely align. In Figures 6(a) and 6(b), Several smaller structures on the right side of the Images show moderate variability. While they maintain similar general positions, their exact boundaries and shapes differ noticeably between the simulations. From Figure 5, a few of the smallest flow structures, particularly some circular formations, show relatively consistent positioning across the different viscosity conditions, suggesting these are more stable flow features less affected by viscosity changes. The positional variability observed in these isocontours demonstrates how sensitive flow structures can be to changes in fluid parameters.

3.

Ans:

Contour boxplots visualize uncertainty in ensembles of contours from simulation data.

Similarities with 1D Boxplots:

Both boxplots provide order statistics such as percentiles, quartiles, median, etc., to summarize data distributions. Both plots help to identify central tendency with medians. They do percentile bands that help to visualize the spread of data. Both of them highlight outliers. Both use the concept of "depth" to order data. In traditional boxplots, the median is the central value and thus has the greatest depth. Other points are ordered based on how far they are from the median, creating a center-outward ordering. In the contour boxplot, this idea is extended using "band depth." Band depth determines how often this contour falls within bands created by other contours. Contours frequently within bands formed by other contours are considered deeper or more central to the distribution. The deepest contour becomes the "median contour". Contours with very low depth are potential outliers.

Key Differences with 1D Boxplots:

1D boxplots work with scalar values, and contour boxplots operate on geometric shapes called contours. Contour boxplots use "band depth" to detect median and other distributions, where a 50% band shows the central half of the distribution, and a 100% band, excluding outliers, shows the full range. Contour boxplots help to visualize the shape characteristics of the data. In traditional box plots, we use lines, whisker plots, and box plots to plot uncertainties, however, the contour plot helps to determine uncertainties through nested regions. Contour plots require computational algorithms to compute set relationships between contours and require computational power and time.

4.

Advantages:
Contour box plots provide clear statistical information about the ensemble data, showing the median contour, percentile regions, and outliers, which gives users quantitative insights rather than just visual impressions. Unlike spaghetti plots display every contour creates chaos in visual whereas contour box plots offer a cleaner visualization, highlighting the most important contours. Contour box plots explicitly identify and display outlier contours, making unusual patterns immediately apparent, whereas in spaghetti plots, outliers blend with other lines. The median contour preserves the actual shape characteristics of typical ensemble members, rather than averaging them in ways that might create unrealistic smoothing.

Disadvantages:
Some detailed information about individual ensemble members is lost due to aggregations. Contour box plots require significantly more computation than spaghetti plots, which involve calculating band depth metrics and complex set operations. They require users to understand new statistical concepts like band depth and contour median, whereas spaghetti plots are relatively well-known to users.

In the example, the contour box plot provides a much clearer summary of the temperature boundary uncertainty compared to the visual overload of the spaghetti plot.

Part 2

1.

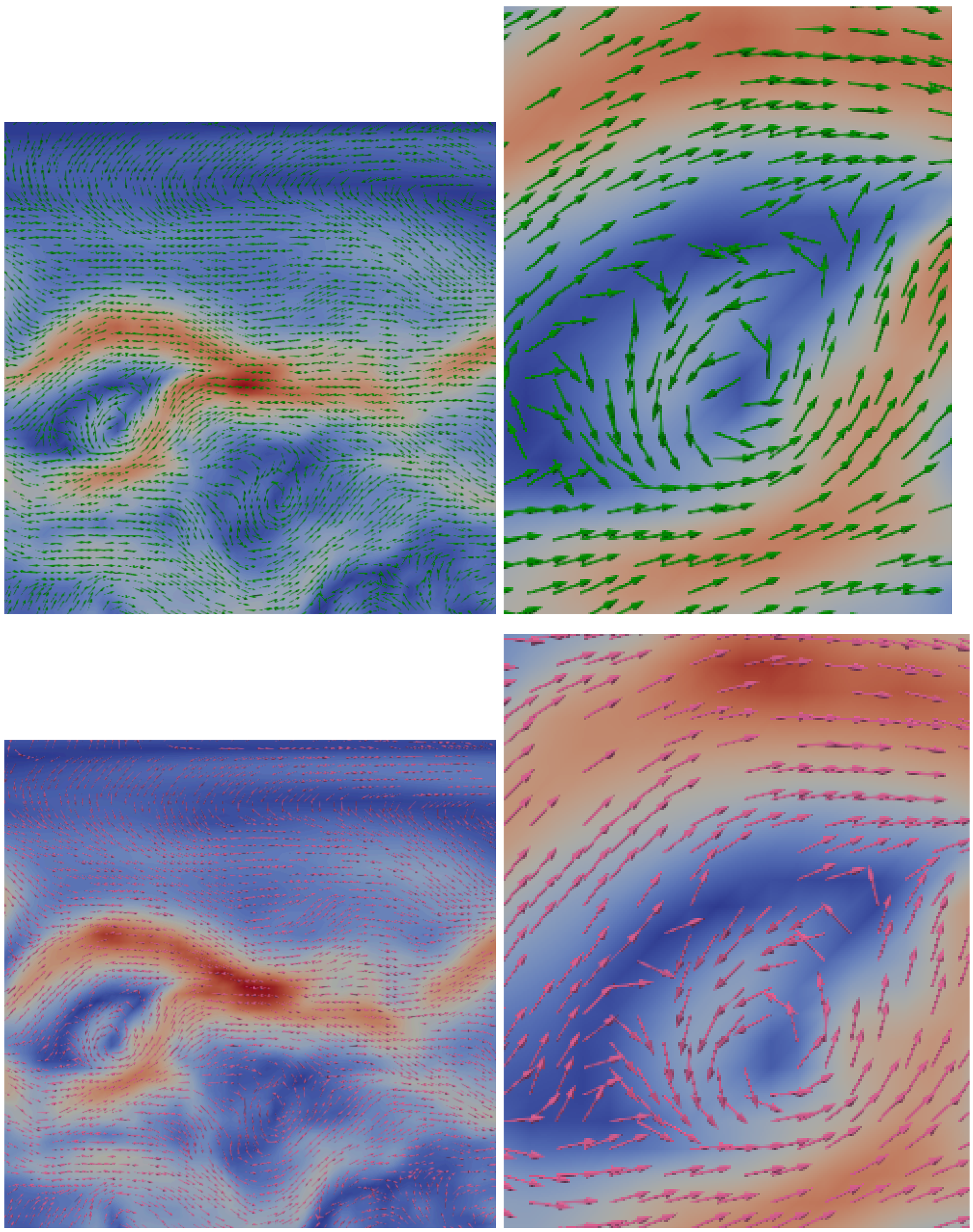


Figure 7: Top left and right: Arrow glyphs with Green color (wind1); Bottom left and right: Arrow glyphs with red color (wind 2)

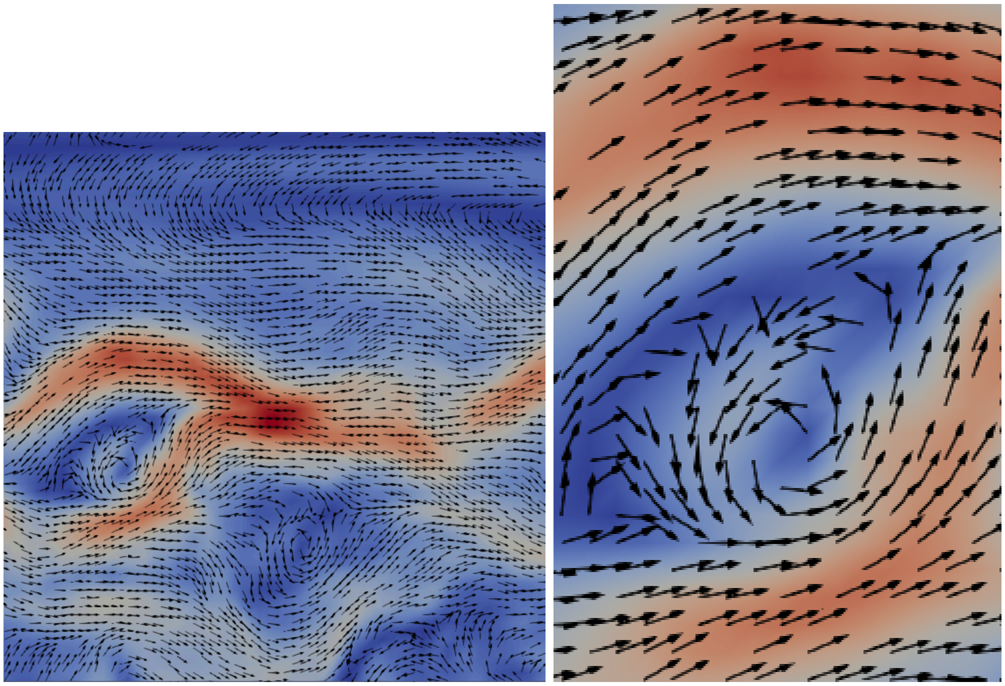


Figure 8: Mean wind with black color glyphs

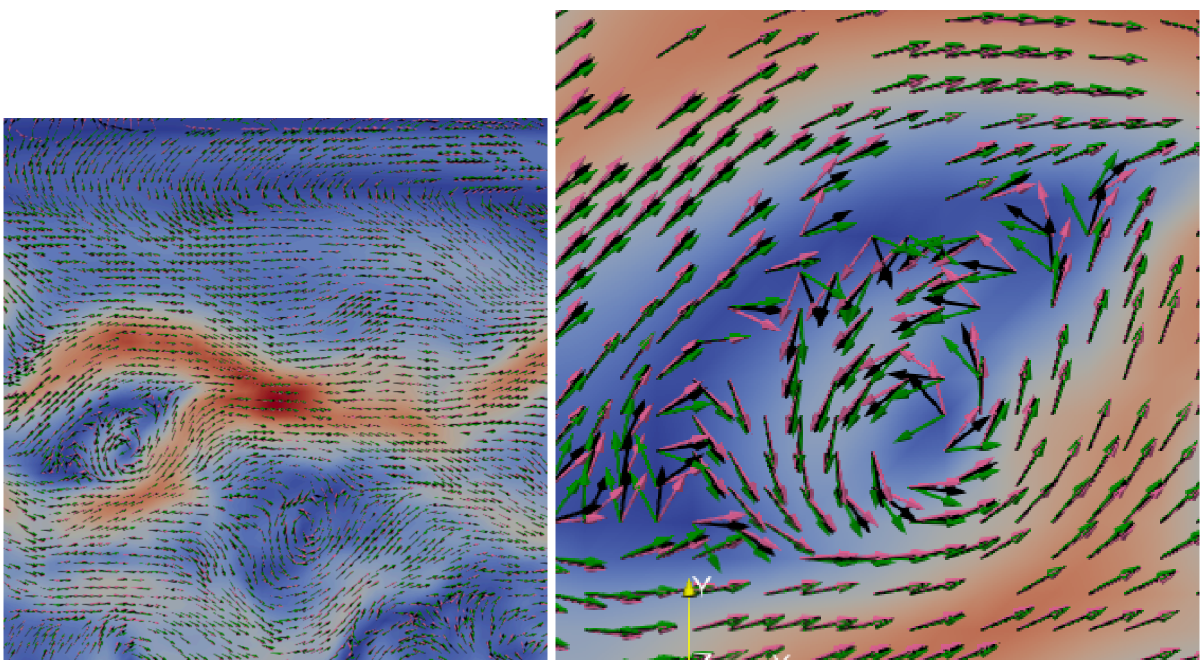


Figure 9: Overlay of the above three images to visualize uncertainty in vector directions.

2.

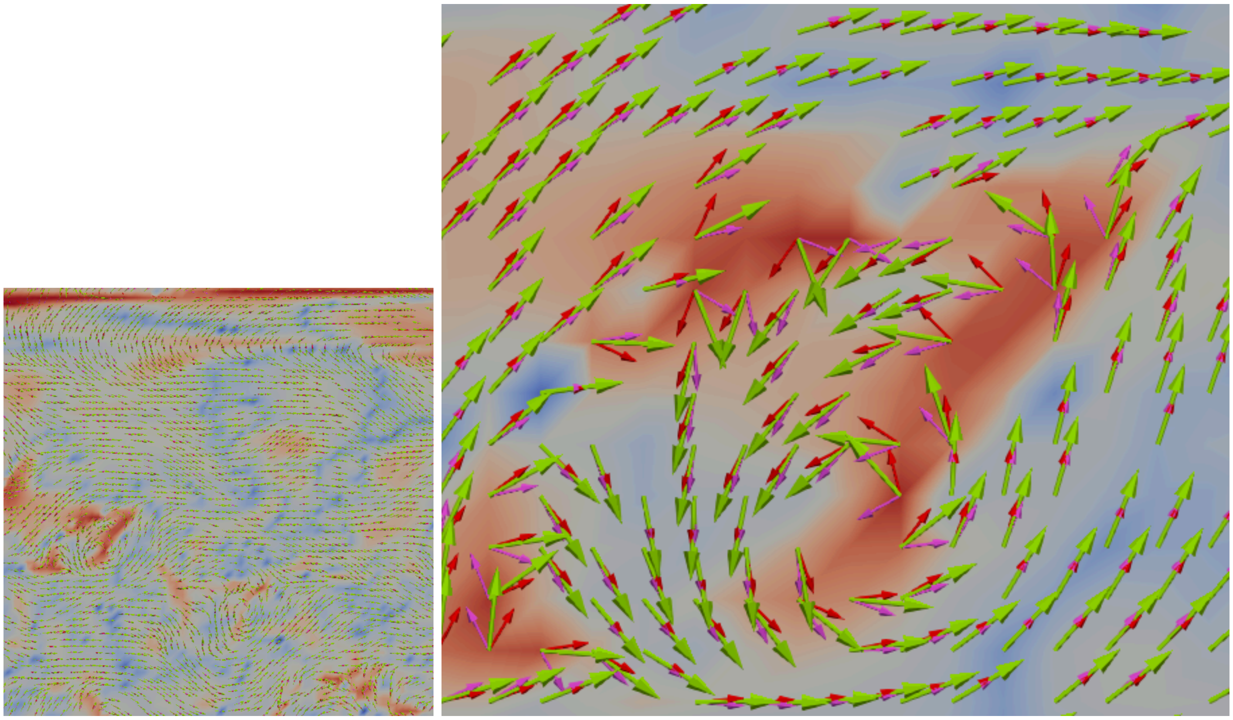


Figure 10: An image of the whole field (left) and a turbulent region (right) with a different color map to visualize the angle between wind 1 and wind 2

Part 3

Q1

Ans: Perceptual channels related to human perceptions take physical stimuli and generate pieces of information. This information is transmitted to the brain. A subset of this information is processed at a very low level in parallel without conscious thought. This term is called preattentive processing. This is necessary in grouping large sets of information, such as visual stimuli. With an understanding of preattentive visual gestures, visualization systems can be designed more effectively and thoughtfully. This processing mechanism is accomplished automatically and simultaneously for the entire visual field of view.

Q2

Ans:
Visual 1:
Form: size, spatial grouping, numerosity

Color: hue and intensity (red has different shades)
Spatial position: 2D position

Visual 2:
Form: size (one shape has different sizes), curvatures different shapes), added marks (circle, triangle, plus sign, inverted triangles)
Color: hue and intensity (red has different shades)
Spatial position: 2D position

Part 4

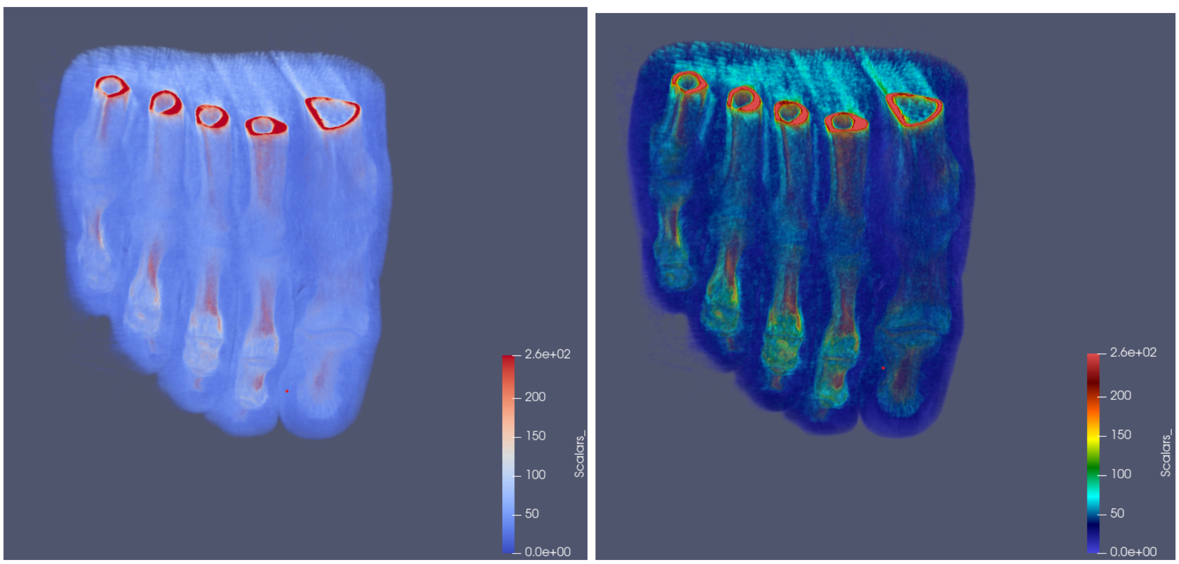


Figure 11: Colormaps provided in Paraview: Cool to Warm (left) and Rainbow Desaturated (right)

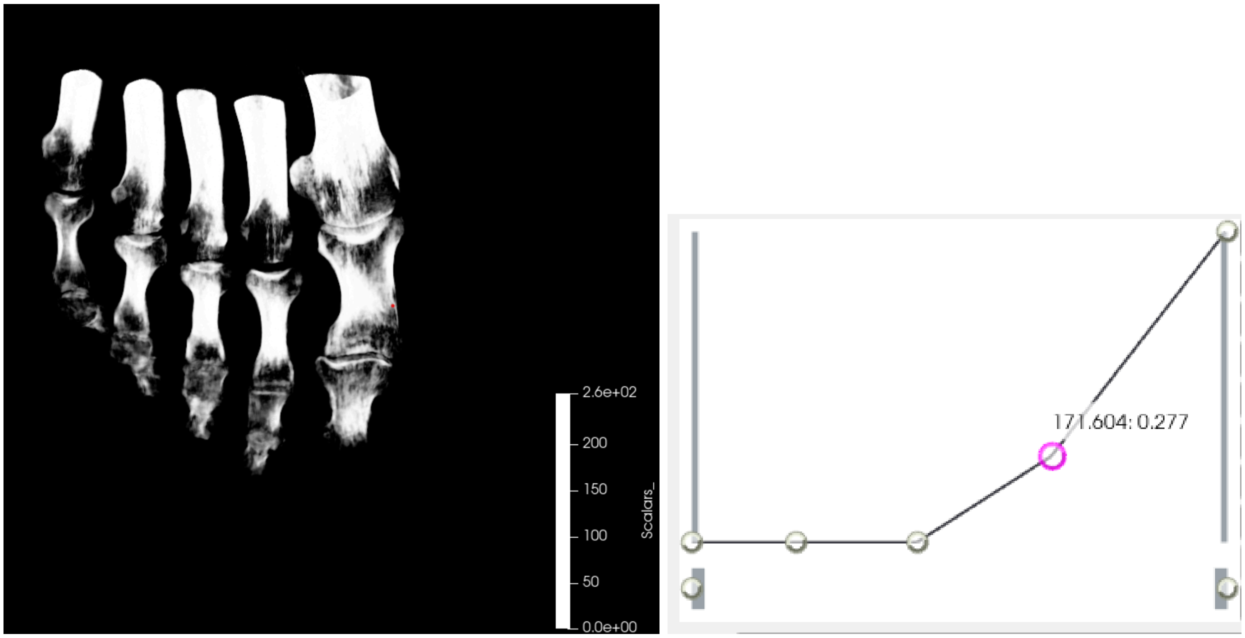


Figure 12: Colormap 1: Black and White colormap transfer function at right and the output image at left

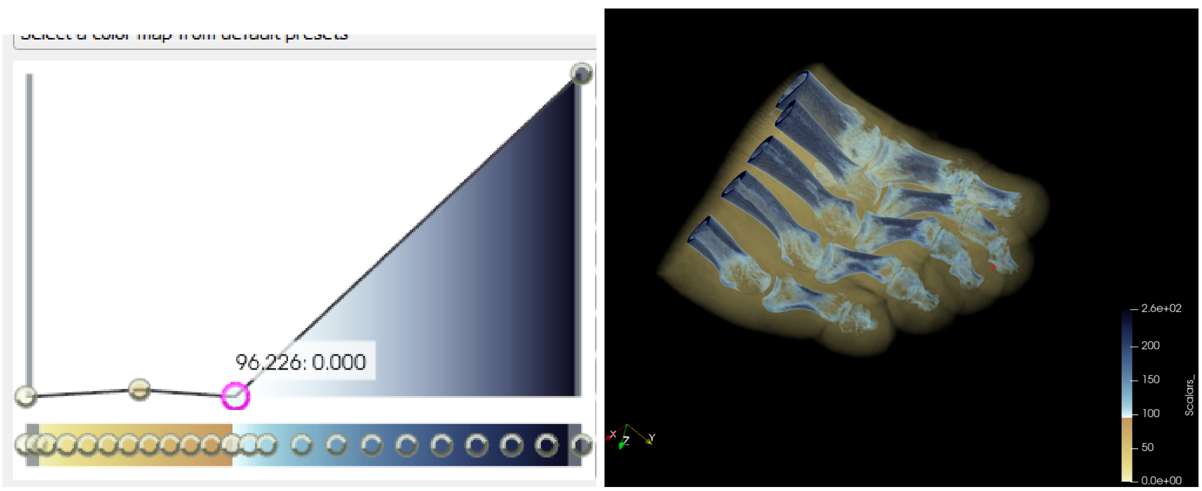


Figure 13: Colormap 2: Left shows the transfer function for the custom color map, and right shows the output image

Conclusion

Part 1 analysis demonstrates the importance of uncertainty visualization and the power of frequency-domain techniques for signal recovery. I learnt lot about contours and their advantages. Part 3 was very intuitive in terms of learning and part 4 took time to generate but I learnt about custom colormaps. Part 2 was very nice in terms of learning the arrow glyphs and angle between two different things.